communications such as fax and SMS, and offer different levels of encryption and security. These protocols are primarily used for voice communication over mobile phones.

GPRS marks the transition into 2.5G. GPRS is a radio technology for GSM networks that adds packet-switching protocols and the possibility for billing by the amount of data sent, rather than connection time. Packet switching is a technique whereby the voice or data is broken up into small chunks of data, which are then routed by the network. EDGE will also be a significant contributor to 2.5G. It is an enhanced modulation technique designed to increase network capacity and data rates. GPRS supports flexible data transmission rates as well as continuous connection to the network, thus making it the most significant step towards 3G.

3G, the Third Generation protocol, is a new network technology being developed for mobile operators. The Universal Mobile Telecommunications System (UMTS) will support much higher data rates, allowing true broadband on mobile devices. 3G networks are already emerging in Japan (NTT DoCoMo is the first 3G network) and some parts of Europe.

2G networks provide about 9.6 Kbps throughput and 2.5G networks increase it to around 56 Kbps. 3G technologies promise upwards of 100-300 Kbps initially, and aim to shoot up to 4 Mbps over time, allowing a range of high-bandwidth multimedia services on mobile devices.

Why GPRS tops

GPRS is built over existing standards, both wired (the Internet) and wireless (mobile communications). Hence, its applications are many and implementation is easy. It's fast, secure, packet-based, and has Quality of Service (QoS) features.

So far, most voice communication happens over circuit-switched networks. Because of this, users have to set up a dedicated connection before they can communicate any information. Hence, when you make a phone call from a landline, a single channel is allocated entirely between you and the receiver. Though this is taken care of by the service providers and is completely transparent, it means a huge waste of resources—the channel is locked between two people even when no data is being transferred. With a packet-based service, more users can be supported over the same channel. The most popular packet-based service is the Internet.

Just like dial-up Internet connections, mobile devices will have an IP address, either static or dynamic. IP packets can originate and travel along the entire network, for example, from your PDA to your e-mail service provider. When you send information over the available channel, the packet contains details such as who it is from and who it is meant for. Thus, several packets can travel on the same channel without getting mixed up and reduce the amount of total bandwidth that is required to support multiple users.

There are several speeds at which GPRS is available. The service providers, depending on how much they invest on the infrastructure, can control this. Theoretically, the maximum rate is around 114 Kbps, though initially we would see much lesser speeds.

QoS, another standard network protocol, can be enabled on GPRS and turns out to be extremely advantageous at low speeds. QoS allows priority allocation to certain types of data packets over others. Thus, applications can define the importance of the data and take up only limited bandwidth, freeing up bandwidth for other applications. For example, you can set e-mail to be received at low speeds and use the rest of the bandwidth for videoconferencing.

GPRS is also being evolved into much faster standards such as Wideband Code Division Multiple Access (W-CDMA), a 3G wireless standard.

How GPRS works

GPRS works over existing GSM networks. Two devices, the Serving GPRS Support Node (SGSN) and Gateway GPRS Support Node (GGSN) convert the circuit-switched network to a packet-switched one. When connecting to the Net, either from the phone, laptop or handheld, the phone sends out signals to the nearest base station. The base station relays the data to the base station controller that communicates with the Mobile Switching Centre. The Mobile Switching Centre connects to the GPRS devices via a Home Locator Register, which tells your account profile to the network. The SGSN tracks your location so that it can keep the communication alive. It also compresses and encrypts the data. The GGSN assigns an IP address to the cell phone and maintains your account logs. It can connect you to other mobile networks just like the roaming service on a cellular phone (*see infographic: The GPRS Network*).

GPRS networks use 200 KHz radio

The Good and the Bad

Wireless Technology	Advantage	Disadvantage
GSM	Widespread; devices easily available and economical	Limited applications; maximum bandwidth 9.6 Kbps
GPRS	Easy to upgrade from GSM; many aplications	Devices expensive; not many available yet
EDGE	Easy to upgrade from GPRS; maximum bandwidth 384 Kbps	Devices not readily available; limited bandwidth currently available
UMTS	Data rates up to 2 Mbps; can grow to 4 Mbps	Service not widespread; still under development

Generations of Wireless Technologies